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COURSE : CSA0927 – PROGRAMMING IN JAVA

Q1 **A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.** 10 marks

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

Write a Java program to:

Create an ArrayList of student names.
Read 8 student names from the user.
Traverse the list using an Iterator.
Display each student name on a separate line.
Display the total number of students stored in the list.

Your Code

```java
import java.util.*;
public class StudentList{
    public static void main(String[] args){
        ArrayList<String>list=new ArrayList<>();
        list.add("Priya");
        list.add("Anu");
        list.add("Kavi");
        list.add("Meena");
        list.add("Vijay");
        list.add("Reena");
        list.add("Divya");
        list.add("Hari");
        Iterator<String>it=list.iterator();
        System.out.println("Student Names");
        while(it.hasNext()){
            System.out.println(it.next());
        }
        System.out.println("Total Stusents="+""+list.size());
    }
}
```

Output

```
Student Name
Priya
Anu
Kavi
Meena
Vijay
Reena
Divya
Hari
Total Stusents=8
```

Q2 **Remove Even Numbers Using Iterator** 10 marks

A company maintains a list of employee ID numbers. The management wants to remove all employees whose ID numbers are even, using an Iterator.

Write a Java program to:

Create an ArrayList<Integer>.
Read 10 employee ID numbers from the user.
Traverse the list using an Iterator.
Remove all even ID numbers using the remove() method of the Iterator.
Display the updated list containing only odd employee IDs.

```java
import java.util.*;
public class RemoveEven {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(101);
        list.add(102);
        list.add(103);
        list.add(104);
        list.add(105);
        list.add(106);
        list.add(107);
        list.add(108);
        list.add(109);
        list.add(110);
        Iterator<Integer>it =list.iterator();
        while (it.hasNext()) {
            int id=it.next();
            if (id%2 == 0)
                it.remove();
        }
        System.out.println("Employee IDs after removing even numbers:");
        System.out.println(list);
    }
}
```

Output

```
Employee IDs after removing even numbers:
[101, 103, 105, 107, 109]

=== Code Execution Successful ===
```

Q3 **Product Price Analysis Using Iterator** 10 marks

A supermarket stores the prices of various products in an ArrayList<Double>. The manager wants to analyze the prices using an Iterator.

Write a Java program to:

Create an ArrayList<Double>.
Read the prices of 10 products from the user.
Traverse the list using an Iterator.
Calculate and display:
Total price of all products.
Average product price.
Highest product price.
Display all product prices using the Iterator.

Your Code

```java
import java.util.*;
public class Product{
    public static void main(String[] args){
        ArrayList<Double>price=new ArrayList<>();
        price.add(100.0);
        price.add(250.0);
        price.add(150.0);
        price.add(300.0);
        price.add(120.0);
        price.add(180.0);
        price.add(220.0);
        price.add(90.0);
        price.add(400.0);
        price.add(160.0);
        Iterator<Double>it=price.iterator();
        double total=0;
        double highest=price.get(0);
        System.out.println("Product Price:");
        while(it.hasNext()){
            double p =it.next();
            System.out.println(p);
            total+=p;
            if(p>highest)
                highest=p;
        }
```

```
        double average=total/price.size();
        System.out.println("Total Price:"+ total);
        System.out.println("Average Price:"+average);
        System.out.println("Highest Price:"+ highest);
    }
}
```

Output

```
Product Price:
100.0
250.0
150.0
300.0
120.0
```

```
180.0
220.0
90.0
400.0
160.0
Total Price:1970.0
```

```
Average Price:197.0
Highest Price:400.0
```